

Experiment No : 08

Experiment Name: Study of inverting and non-inverting amplifier using Op-Amp

Objectives:

- To implement inverting and non-inverting amplifier using Op-Amp
- To observe the input/output waveforms of inverting and non-inverting amplifier

Theory:

□ Inverting Amplifier:

An inverting amplifier is an Op-Amp configuration where the input signal is applied to the inverting terminal (-) and the non-inverting terminal (+) is grounded...

□ Non-Inverting Amplifier:

A non-inverting amplifier applies the input signal to the non-inverting terminal (+) and the inverting terminal (-) is connected to a feedback network...

The gain of inverting and non-inverting amplifier is provided by,

$$V_o = - \frac{R_f}{R_1} \times V_i$$

Inverting

and

$$V_o = \left(1 + \frac{R_f}{R_1}\right) \times V_i$$

Non-inverting

Required Apparatus:

- (i) Op-Amp (u741, 1 piece)
- (ii) Resistors ($1\text{ k}\Omega$, $10\text{ k}\Omega$: 1 piece each)
- (iii) Signal Generator
- (iv) Oscilloscope
- (v) DC power supply (0-15 V)
- (vi) Project Board
- (vii) Multimeter
- (viii) Connecting wires

Circuit Diagram:

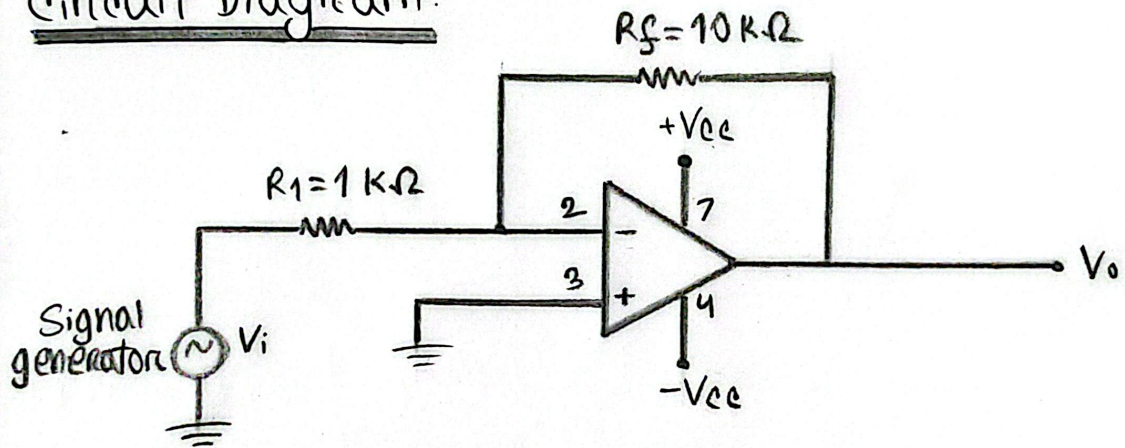


Figure: Inverting Amplifier

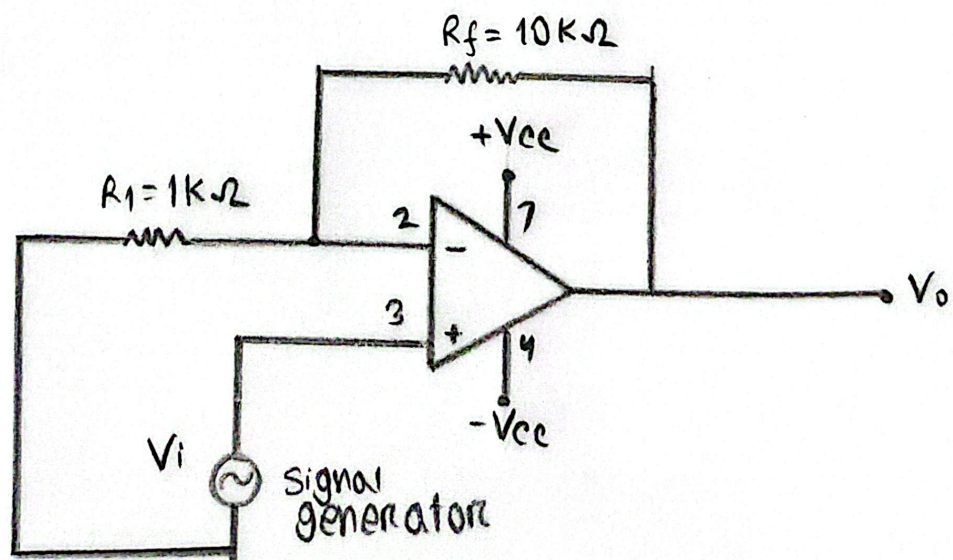


Figure: Non-Inverting Amplifier

Procedure:

- (i) The circuit was connected according to figure(a)
- (ii) An AC signal supply was given from the signal generator and the corresponding output waveform was recorded
- (iii) Comparison between the measured gain and theoretical gain was made
- (iv) Again, the circuit was connected according to figure(b)
- (v) Steps (ii) and (iii) were repeated

Data Table:

	Input Voltage, $V_{in}(P-P)$ (V)	Output Voltage, $V_{out}(P-P)$ (V)	Measured Gain, $(A_v = V_o/V_i)$	Theoretical Gain, (A_v)	% of Error
Inverting Amplifier	2.040	-20.40	10	10	0
Non-Inverting Amplifier	2.120	23.60	11.132	11	1.2

Discussion and Conclusion:

In this experiment, the inverting and non-inverting amplifiers were successfully implemented and the output waveforms agreed with theoretical expectations. The measured gains showed minimal errors (0% and 1.2%) confirming proper circuit operation.

Precautions:

- All the connections were given properly
- Op-Amp's terminals were identified correctly
- Calibration were performed properly